

How Well Do Frontier LLMs Evaluate Research? Evidence from Structured Expert Peer Review

Valentin Klotzbücher David Reinstein Lorenzo Pacchiardi
Tianmai Michael Zhang

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Abstract

We evaluate how well frontier large language models (LLMs) can critique, rate, and prioritise research when constrained to the same structured rubric used in paid expert peer review. Using published evaluation packages from The Unjournal, we compare LLM-generated quantitative ratings (percentile scores with 90% credible intervals across seven criteria) and qualitative critiques to expert assessments for economics and social-science working papers. Human evaluations serve as a high-quality but noisy reference signal, allowing us to assess agreement while accounting for substantial inter-rater variability. Across models, we find moderate agreement with expert mean ratings, with the strongest-performing reasoning models approaching—though not reaching—the ceiling implied by human–human agreement. A consistent pattern is central compression: LLMs shrink extreme ratings toward the middle and often report narrower uncertainty than experts. On qualitative critique alignment, LLMs sometimes capture much of the expert-identified concern set but performance varies widely by paper, with frequent omissions of key issues and occasional emphasis on concerns not prioritised in expert syntheses. Cost differences across models are large, suggesting a practical tiered use: inexpensive models for rapid screening and stronger reasoners for deeper assessment, while retaining human oversight for high-stakes decisions. Overall, frontier LLMs can provide fast, structured evaluative signals, but systematic calibration and coverage gaps argue against fully automated peer review.

Introduction

Working paper

This is an evolving working paper. Analysis, metrics, and comparisons are under active development.

Peer review is under strain. Reviewers are hard to find, turnaround times are lengthening, and the system costs an estimated \$1.5 billion per year in the United States alone (Aczel, Szaszi, and Holcombe 2021). An influx of AI-generated manuscripts is likely to intensify the bottleneck (Eger et al. 2025). If frontier language models could produce reliable, structured evaluations of research, they might free substantial scientific resources and deliver faster, more detailed feedback.

We test this possibility using The Unjournal’s structured human evaluations as a benchmark. We prompt six frontier LLMs—GPT-5 Pro, GPT-5.2 Pro, GPT-4o-mini, Claude Sonnet 4, Claude Opus 4.6, and Gemini 2.0 Flash—with the same rubric and guidelines used by human evaluators, then compare the resulting quantitative ratings and qualitative critiques against human evaluations for 60 economics and social-science working papers. We ask whether frontier AI evaluations can

reliably approximate human expert judgment, which models perform best across rating criteria and cost tiers, and whether systematic differences reveal characteristic AI preferences over research.

The Unjournal setting is particularly well suited for this comparison. It commissions paid expert evaluations using a structured rubric covering seven percentile criteria with 90% credible intervals plus journal-tier predictions, and publishes the resulting packages openly—reducing classic gatekeeping motives and increasing reviewer effort. The resulting ratings and critiques still exhibit substantial inter-rater variation; accordingly, we treat human evaluations as a high-quality but noisy *reference signal*, not ground truth. The rich, multi-dimensional data allow us to compare the priorities and calibration of humans and AI models across criteria and domains, while the journal-tier predictions provide an external reference point¹ enabling a human-vs-LLM horse race. Finally, The Unjournal’s pipeline of future evaluations allows for clean out-of-training-data predictions, serving as a live testing lab for prospective validation.

A growing literature examines LLM-assisted peer review, error detection, and AI research evaluation; we survey this work in Appendix C. Our approach differs from prior work by focusing on structured, percentile-based quantitative ratings with credible intervals, comparing against published human evaluations rather than using LLM-as-judge, and curating contamination-aware inputs.

Acknowledgements.

This work is a collaboration with [The Unjournal](#), which has published 55+ detailed evaluation packages building open, quantitative evaluation infrastructure for global-priorities-relevant research in economics, policy, and social science. We thank The Unjournal’s evaluation community for generating the structured human assessments that make this benchmarking possible.

Funding for The Unjournal has been provided by the Survival and Flourishing Fund, the Long Term Future Fund, and EA Funds. Portions of this manuscript were drafted with AI assistance and reviewed by the authors.

Results

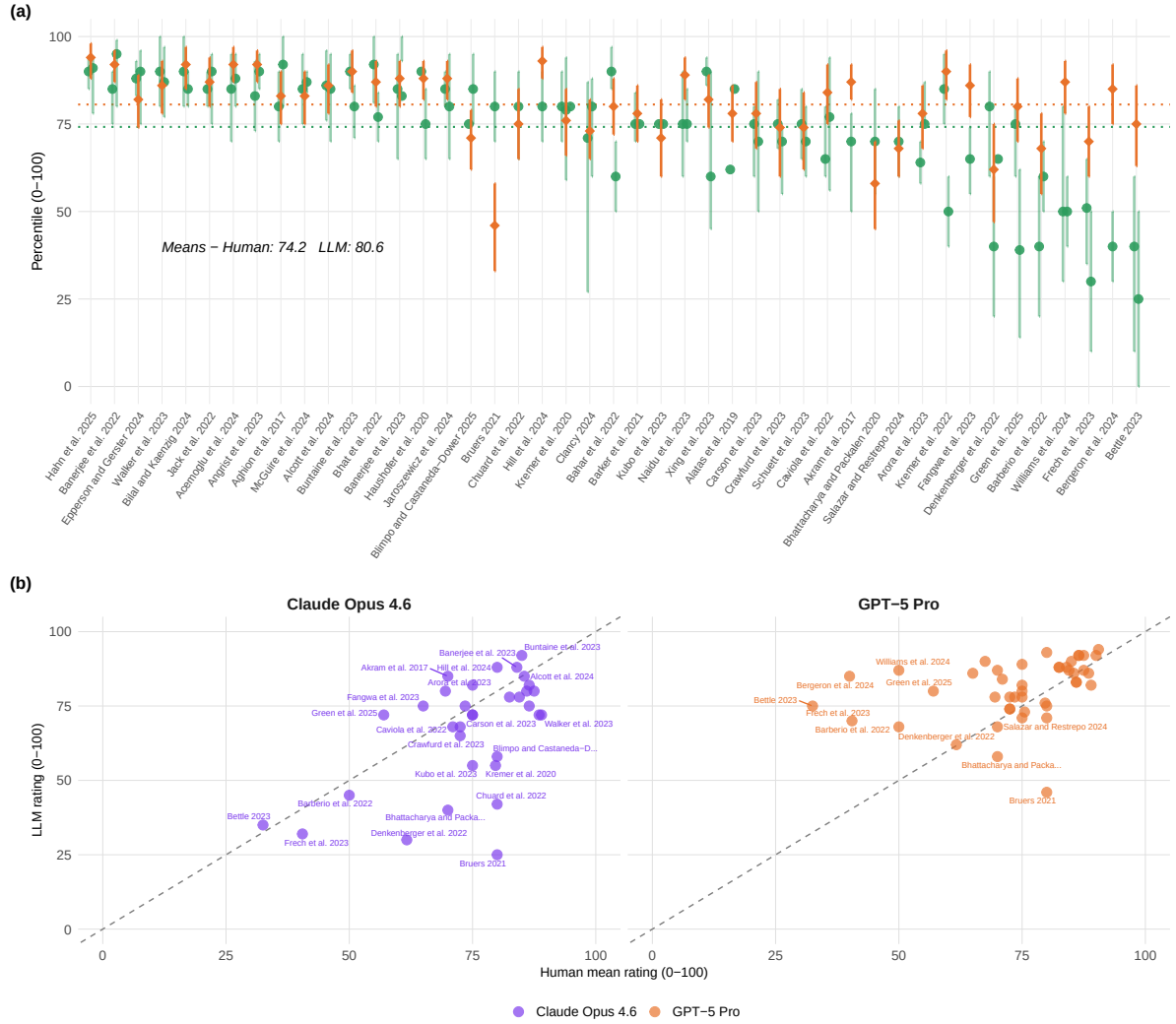
We evaluate 6 frontier LLMs against human expert reviews from [The Unjournal](#) for 45 papers with both human and LLM evaluations. Each model receives the same PDF, system prompt mirroring The Unjournal rubric, and JSON schema requiring a diagnostic summary plus numeric midpoints and 90% credible intervals for every metric. Full methodological details appear in [Methods](#).

Per-paper overview and model comparison. Figure 1 presents two complementary views of overall (0–100 percentile) ratings. Panel (a) displays individual human evaluator ratings alongside the GPT-5 Pro LLM rating for each paper, revealing inter-rater variability—CIs often span 20–40-points. In most cases the LLM falls within the range of human opinions, though several papers show substantial divergence. Panel (b) compares the two highest-performing reasoning models directly against human mean ratings; points on the diagonal indicate perfect agreement.

Both models cluster around the identity line in the 40–80 range but diverge more at the extremes. LLMs tend to compress ratings toward the centre of the scale relative to humans, producing narrower spread—a pattern consistent with alignment training that discourages extreme outputs. Where humans rate a paper very highly or very harshly, the LLM typically pulls toward the middle. Full agreement metrics including Pearson r , Spearman ρ , mean bias, and RMSE for all six models appear in the [agreement table in Appendix A](#).

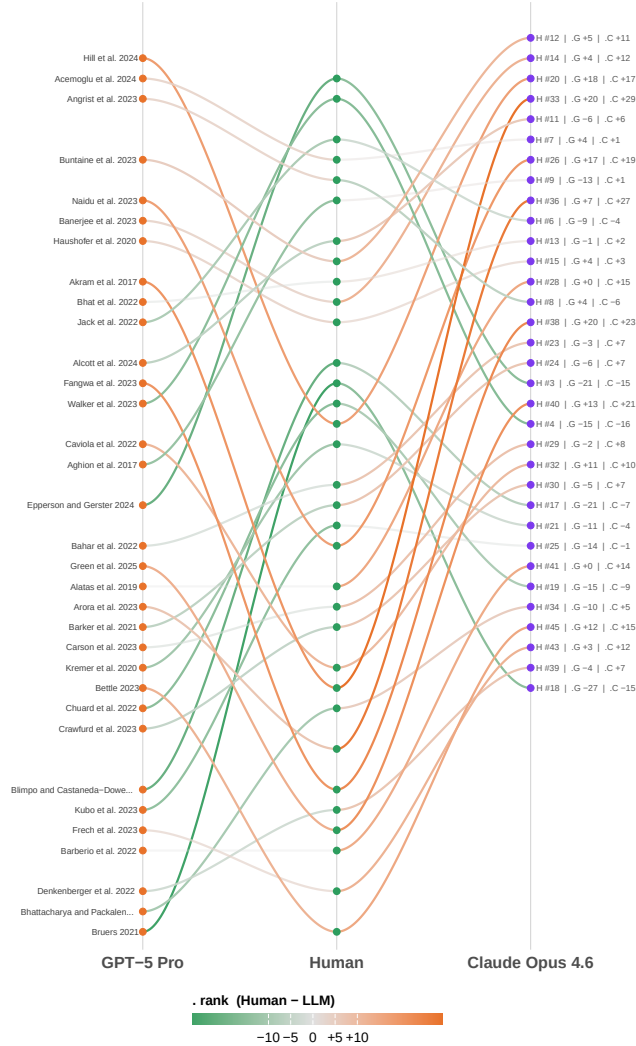
¹These represent verifiable publication outcomes, not statements about the “true quality” of the paper.

Figure 1: Per-paper overall ratings (0–100 percentile). **(a)** Individual human evaluator midpoints (green circles) with 90% credible intervals and GPT-5 Pro rating (orange diamond with CI) for each paper, sorted by descending human mean. Dotted lines mark grand means. **(b)** Human mean vs LLM overall rating for GPT-5 Pro and Claude Opus 4.6. Dashed diagonal is the identity line (perfect agreement); points above it indicate the LLM rated the paper higher.



Rank comparison. Figure 2 places human rankings in the centre column with GPT-5 Pro on the left and Claude Opus 4.6 on the right. Each paper appears as a pair of S-curves connecting its human rank to its rank under each LLM. Horizontal curves indicate agreement; curves that climb toward the top indicate the LLM ranked the paper higher (better) than humans did. Colour encodes the rank shift: green when humans rank it higher, orange/purple when the LLM does. The triptych layout makes it easy to spot papers where both LLMs agree with each other but disagree with humans (both curves slope the same way), versus papers where the two LLMs diverge.

Figure 2: Triptych slopegraph of paper rankings (overall criterion). Centre column shows human mean ranking; left and right columns show GPT-5 Pro and Claude Opus 4.6 rankings respectively. Each paper is connected by S-shaped curves; colour intensity encodes the magnitude of rank shift (Δrank). Green = humans rank it higher; warm colour = LLM ranks it higher. Horizontal curves indicate agreement.

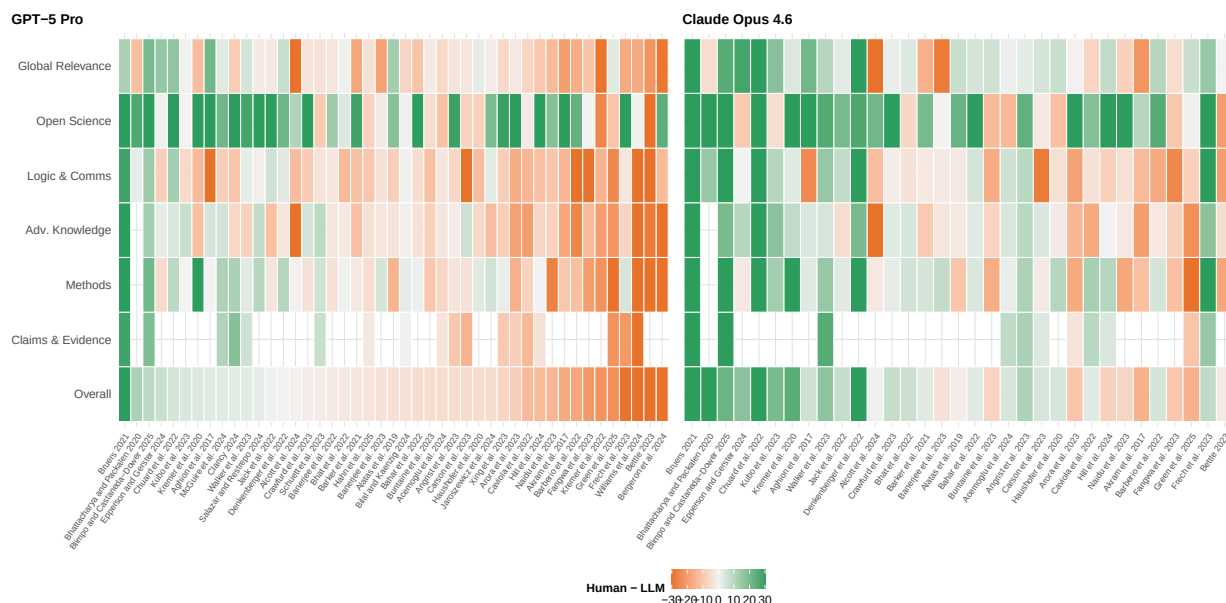


Papers where both curves slope the same direction indicate systematic human–LLM disagreement; papers where the two curves diverge indicate model-specific “taste” differences. The criteria-level Krippendorff’s alpha analysis in the [human baseline table in Appendix A](#) (a chance-corrected reliability coefficient that generalises across varying numbers of raters, missing data, and any measurement level; see [Methods](#) for a formal definition) quantifies where human–LLM agreement ap-

proaches the ceiling set by human inter-rater variability.

Rating differences by paper and criterion. While the scatter plots and slopegraph summarise agreement on a single dimension, Figure 3 unpacks Human – LLM differences across all seven criteria simultaneously. Each column is a paper, each row a criterion, and tile colour encodes the signed difference (human mean minus LLM midpoint). Green tiles indicate the human panel rated the paper higher; orange tiles indicate the LLM was more generous. Papers are sorted by GPT-5 Pro’s overall difference so that both panels share the same column order, making model-specific patterns immediately visible.

Figure 3: Human minus LLM rating difference for every paper (columns) and criterion (rows), comparing GPT-5 Pro (left panel) and Claude Opus 4.6 (right panel). Green tiles indicate the human mean was higher; orange tiles indicate the LLM rated the paper higher. Colour is clamped to ± 30 percentile points. Papers are sorted by GPT-5 Pro overall difference so the two panels are directly comparable.



Rows with uniformly green or orange tiles indicate papers where humans and the LLM disagree systematically across all criteria, not just on one dimension. Columns with consistent colour suggest criteria-level biases—for instance, if “Open Science” is green across nearly all papers, humans may systematically reward data-sharing practices more than the LLM does. Comparing the two panels reveals whether disagreement patterns are model-specific or shared across frontier LLMs; tiles that change colour between panels highlight papers where the models diverge from each other, not just from humans.

Qualitative critique comparison. Beyond numeric ratings, we compare the substantive critiques each model raises against the consensus issues identified by human experts. The figure below plots coverage—the fraction of human-identified concerns that the LLM also raised in some form—against precision—the fraction of LLM-raised issues that correspond to a substantive human concern. These metrics are assessed by GPT-5.2 Pro acting as an independent judge (see [Methods](#) for the judging protocol).

Coverage and precision vary substantially across papers. Some papers achieve high scores on both dimensions, indicating strong alignment between human and LLM critiques, while others reveal the

LLM missing key human concerns or raising issues absent from the expert consensus. Because these metrics are themselves LLM-assessed, they should be interpreted with the caveat that an LLM judge may systematically over- or under-credit matches relative to a human annotator; manual validation through our annotation tool is underway and preliminary results are consistent with the automated scores.

Detailed paper-by-paper comparisons—including matched issue pairs with severity labels, structural difference tables, and per-evaluator breakdowns—appear in [Appendix B: Critiques & Key Issues](#). Extended quantitative analysis, including per-criterion correlations, bootstrap confidence intervals, mutual information, tier prediction accuracy, cost-quality trade-offs across all six models, and top disagreement cases, is reported in [Appendix A: Results Ratings](#). The full LLM reasoning traces and assessment summaries are available in [Appendix C: LLM Traces](#).

All appendices, interactive figures, and full code are available at the companion website: <https://llm-uj-research-eval.netlify.app>. Source code and data are hosted at <https://github.com/valentinklotzbuecher/llm-uj-research-eval>.

Discussion

Frontier LLMs achieve moderate-to-strong agreement with human expert evaluators, approaching the ceiling set by inter-rater variability among humans themselves. Correlations with human mean ratings are highest for reasoning-capable models (GPT-5 Pro, Claude Opus 4.6), though even lightweight models (GPT-4o-mini, Gemini 2.0 Flash) perform respectably at a fraction of the cost. Central compression—the tendency for LLMs to pull extreme ratings toward the middle of the scale—is the most consistent pattern across all models, likely reflecting alignment training that discourages confident extreme outputs. Qualitative coverage varies widely across papers: on some, the LLM captures nearly all consensus human concerns; on others, it misses key critiques or raises issues absent from the expert consensus.

Limitations. Several caveats temper these conclusions. Our sample comprises roughly 50 social-science papers specifically selected by The Unjournal for evaluation, not a random draw from the research literature; performance may differ in other fields or on less polished manuscripts. Human evaluations are themselves a noisy reference signal rather than ground truth, with substantial inter-rater variation that caps achievable agreement. We cannot fully rule out knowledge contamination: while we block extrinsic priors in the system prompt, the models’ training data may include fragments of these papers or related discussions. Alignment training likely contributes to score inflation and narrower credible intervals than humans provide. All LLM evaluations are single-run; aggregating across multiple runs or temperature settings could change the picture. Finally, the qualitative coverage and precision metrics are themselves LLM-assessed (GPT-5.2 Pro as judge), introducing a further layer of model dependence.

Implications. The cost structure is striking: lightweight models deliver evaluations at under a cent per paper, while reasoning-capable models cost several dollars—still orders of magnitude cheaper than human expert review. This suggests a practical tier structure in which cheap models could serve as rapid screening tools and expensive reasoners could provide deeper assessment where warranted. However, the qualitative gaps we observe—missed critiques, generic issues, and central compression of ratings—argue against full automation of peer review. AI evaluation appears most promising as a supplement: providing fast structured feedback, flagging potential concerns for human reviewers, and enabling systematic comparison across large paper sets that would be infeasible with human effort alone.

Future work. Several extensions follow naturally from this analysis. Content-swap bias tests following Pataranutaporn et al. (2025) would reveal whether LLMs show systematic biases based on author names or institutional affiliations. A journal-outcome prediction horse-race would compare human and LLM tier predictions against actual publication venues. Systematic prompt and model comparisons, including agent-based approaches, could identify configurations that reduce central compression or improve qualitative coverage. Human enumerator validation—employing trained raters to independently assess whether LLMs identify the same consensus critiques—would ground the currently LLM-assessed coverage metrics. Out-of-time validation using papers entering The Unjournal’s pipeline would eliminate contamination concerns entirely. Finally, hybrid human-AI evaluation trials would test whether collaboration improves evaluation quality beyond either alone.

Methods

This chapter describes the data sources, evaluation pipeline, prompt design, and statistical methods used throughout. All Python code underlying the pipeline is preserved in the collapsible blocks below and can be executed to reproduce the evaluation runs.

Sample and human reference data. We draw on published evaluations from [The Unjournal](#), an open-access platform that commissions expert reviews of policy-relevant research without requiring journal submission. Each paper is typically assessed by two independent evaluators (occasionally one or three), who provide a written critique resembling a referee report together with quantitative ratings on seven criteria scored as percentiles (0–100) relative to “all serious research in the same area encountered in the last three years.”² Evaluators additionally predict the journal tier in which the work “should” and “will” publish, using a 0–5 continuous scale anchored to familiar venue categories (0 = unpublishable, 5 = top-5 journal). For each metric, evaluators report a midpoint (median of their belief distribution) and a 90% credible interval that expresses their epistemic uncertainty.

The sample comprises working papers spanning 2017–2025 in development economics, health policy, environmental economics, and related fields that The Unjournal identified as high-impact. All papers have completed The Unjournal’s full evaluation process, meaning the authors received evaluations that have been publicly posted. For our analysis we extracted the individual evaluator scores, aggregated them by taking the arithmetic mean per paper per criterion, and retained the range of individual scores to characterise inter-rater spread.

LLM evaluation pipeline. We evaluate six frontier models spanning three providers: GPT-5 Pro and GPT-5.2 Pro (OpenAI, reasoning-capable), GPT-4o-mini (OpenAI, lightweight), Claude Sonnet 4 and Claude Opus 4.6 (Anthropic), and Gemini 2.0 Flash (Google). Each model receives the identical PDF file, system prompt, and output schema. We pass the PDF directly to the model’s native multimodal input rather than extracting text, preserving tables, figures, equations, and layout cues that ad-hoc scraping could mangle. A single API call per paper avoids hand-offs and summary loss from multi-stage pipelines.

The output is constrained by a strict JSON Schema enforcing the same nine fields that human evaluators complete: seven percentile metrics (each with `midpoint`, `lower_bound`, `upper_bound`) and two journal-tier predictions (each with `score`, `ci_lower`, `ci_upper`). Additionally, the model produces an `assessment_summary` of approximately 1,000 words that must precede scoring—a “think first, score second” protocol designed to ground numeric ratings in specific textual evidence. The extended schema used for our GPT-5.2 Pro focal run adds a `key_issues` array of concise, ranked issue statements for downstream critique comparison.

Agreement and reliability statistics. We quantify human–LLM agreement using several complementary measures. *Pearson’s r* captures linear association between paired ratings and is sensitive to proportional biases but not to constant offsets. *Spearman’s* measures rank-order agreement and is robust to outliers and non-linear monotone relationships. *Mean bias* (LLM minus human) indicates the direction and magnitude

²The seven criteria are: *overall assessment, claims and evidence, methods, advancing knowledge and practice, logic and communication, open and collaborative science, and relevance to global priorities*. Full definitions are given in The Unjournal’s [guidelines for evaluators](#).

of any systematic offset. *Root mean squared error* (RMSE) and *mean absolute error* (MAE) measure typical prediction error in the original scale units; RMSE penalises large deviations more heavily. To contextualise human–LLM agreement we report *Krippendorff’s alpha* (α), a chance-corrected reliability coefficient that generalises across varying numbers of raters, accommodates missing data, and applies to any measurement level (nominal, ordinal, interval, or ratio). An α of 1 indicates perfect agreement, 0 indicates agreement no better than chance, and values below 0 indicate systematic disagreement. We compute α_{HH} (among human evaluators only) as a ceiling: if human evaluators agree with each other at only $\alpha = 0.5$ on a given criterion, expecting an LLM to exceed that level would be unrealistic. We then report α_{HL} (between the human mean and each LLM) to assess how close machine ratings come to this ceiling. For the qualitative key-issue comparison, *coverage* denotes the fraction of human-identified issues that received a match score $\geq 30\%$ from the LLM judge, and *precision* denotes the fraction of LLM-generated issues that matched at least one human issue.

JSON Schema. The output schema enforces that every paper is scored on identical fields with identical types and bounds. Credible intervals are required (paralleling the human protocol) so that the model can express genuine uncertainty rather than suggest false precision.

System prompt design. The system prompt is assembled from modular components and concatenated before each API call. It opens with a role definition instructing the model to act as an expert research evaluator, followed by a debiasing block that explicitly prohibits use of author identity, institutional prestige, publication venue, or any extrinsic information—the model must base all judgments on the PDF content alone. A diagnostic-summary instruction requires the model to produce a roughly 1,000-word assessment identifying methodological, evidential, and interpretive issues *before* any scoring, implementing a “think first, score second” protocol intended to anchor numeric ratings in specific textual evidence.

Percentile ratings are anchored to the reference group “all serious research in the same area encountered in the last three years,” following The Unjournal’s [guidelines for evaluators](#). The prompt defines each of the seven criteria with emphasis on global priorities and practical relevance over pure academic novelty, mirroring the weight structure that human Unjournal evaluators are asked to apply.

Subsequent prompt components instruct the model on constructing 90% credible intervals—the smallest interval the evaluator believes is 90% likely to contain the true value—encouraging calibrated uncertainty rather than artificially narrow bounds. The prompt requests journal-tier predictions on a 0–5 continuous scale anchored to familiar venue categories (0 = unpublishable through 5 = top-5 journal), providing an externally verifiable benchmark for papers that eventually publish. A validation block then requires the model to verify internal consistency: numeric scores must align with the written assessment, credible intervals must be non-degenerate, and high or low ratings must be explicitly justified in the assessment summary.

The components above are concatenated into a single system prompt string before each API call:

Submission and collection. The `evaluate_paper` function uploads a PDF to the API and submits a background job for evaluation. File IDs are cached by path, size, and modification time so that re-running on the same PDF reuses the previously uploaded file.

Each model receives the full PDF in a single reasoning call, avoiding hand-offs and summary loss from multi-stage ingestion. We submit one background job per paper to the OpenAI Responses API with “high” reasoning effort and server-side JSON-Schema enforcement, recording the response ID, model, file ID, status, and timestamps in a jobs index. No external sources or cross-paper material are retrieved; the evaluation is anchored entirely in the manuscript itself.

A polling loop then checks each job’s status and, for completed jobs, retrieves the raw JSON response object and writes it to disk alongside reasoning-trace metadata (token counts, reasoning summary).

Multi-model evaluation. To assess whether systematic biases or calibration differences emerge across architectures, we collect evaluations from six models spanning three providers. For OpenAI models that lack background-job or extended-reasoning support (e.g., GPT-4o-mini), we use a synchronous call variant:

Anthropic. Anthropic’s API accepts PDFs as base64-encoded document content rather than uploaded file

IDs, and all calls are synchronous. We use Claude’s native PDF support to preserve the same multimodal evaluation approach:

Google. Google’s Gemini API accepts PDFs via a file-upload endpoint with MIME-type tagging:

A unified runner dispatches evaluations across all configured providers and writes per-model output directories:

Focal run with key-issue extraction. For a subset of 14 papers with rich human critiques, we ran an extended evaluation using GPT-5.2 Pro. The schema for this focal run adds a `key_issues` array—a ranked list of concise issue statements identifying the most important methodological, interpretive, or evidential concerns—alongside the standard metrics. This structured output enables direct comparison between machine-generated and human-identified issues.

Key-issue comparison with human critiques. To validate how well the LLM identifies substantive concerns, we compare its `key_issues` output against human expert critiques drawn from The Unjournal’s Coda database. These human critiques—produced by paid domain experts and synthesized by evaluation managers—provide a high-quality but noisy reference standard for issue identification; accordingly, we treat them as a comparative signal rather than ground truth.

The comparison proceeds in two stages. First, we parse and align the data sources: LLM key issues are extracted from the focal-run JSON responses, while human critiques are drawn from a manually curated markdown document pairing each paper’s machine and expert assessments. Second, we use an LLM judge (GPT-5.2 Pro with schema-enforced structured output) to systematically assess the degree of alignment between each human-identified issue and the set of machine-generated issues, producing issue-by-issue match scores, coverage estimates (fraction of human issues captured), and precision estimates (fraction of LLM issues that are genuinely substantive).

The comparison pipeline takes a manually curated markdown file pairing each paper’s LLM issues with human critiques and produces two structured outputs: a parsed JSON dataset (`key_issues_comparison.json`) and an LLM-assessed alignment report (`key_issues_comparison_results.json`) containing per-paper coverage, precision, matched-pair explanations, and an overall quality rating. Together, these outputs feed the qualitative analysis presented in [Appendix B: Critiques & Key Issues](#).

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Results: Ratings

Extended ratings analysis (per-criterion correlations, agreement metrics, human baseline, cost-quality trade-offs) is available in the online supplement at https://llm-uj-research-eval.netlify.app/results_ratings.html.

Results: Critiques & Key Issues

Detailed paper-by-paper critique comparisons (matched issue pairs, severity labels, structural differences) are available in the online supplement at https://llm-uj-research-eval.netlify.app/results_critiques.html.

Supplementary Material

Full LLM reasoning traces, assessment summaries, related work, and extended analyses are available in the online supplement at https://llm-uj-research-eval.netlify.app/appendix_llm_traces.html.